



△ **Ethiopian opal** displaying a light-base, full spectrum play-of-colour

Opal falls into two categories: precious and common. The former displays highly prized rainbow iridescence with a white to dark body colour, while the latter has a strong, attractive body colour and no iridescence. Both kinds consist of hardened silica gel, and usually contain 5–10 per cent water in submicroscopic pores. Precious opal consists of a regular arrangement of tiny, transparent, silica spheres, and its colour play occurs when the spheres are regularly arranged and of the correct size, causing the diffraction of light and its consequent break-up into the colours of the spectrum: the actual colours that appear depend on the size of the spheres. Opal is deposited at low temperatures from silica-bearing waters, usually in sedimentary rocks. In ancient times, the primary source was in present-day Slovakia; more recently, Australia was the chief producer, and is also the source of fossil bones and seashells that have been replaced by precious opal. Ethiopia is now the main source of gem opal.

Common opal

Mineralogically, common opal refers to fire opals, which are transparent to translucent and do not usually show a play of colour; it can also refer to opals with no colour or transparency and no gemstone value. Fire opals, sometimes called jelly opals, are prized for their rich colours: yellow, orange, orange-yellow, or red. Transparent fire opals tend to be faceted, and are often set into moderately expensive silver jewellery.

Specification

Chemical name Hydrous silicon dioxide | **Formula** $\text{SiO}_2 \cdot n\text{H}_2\text{O}$
Colours Colourless, white, yellow, orange, rose-red, black, dark blue | **Structure** Amorphous | **Hardness** 5–6 | **SG** 1.9–2.5 | **RI** 1.37–1.52 | **Lustre** Vitreous | **Streak** White



Round brilliant

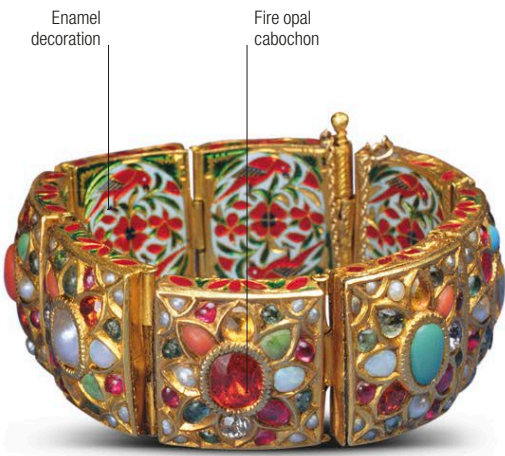


Cabochon

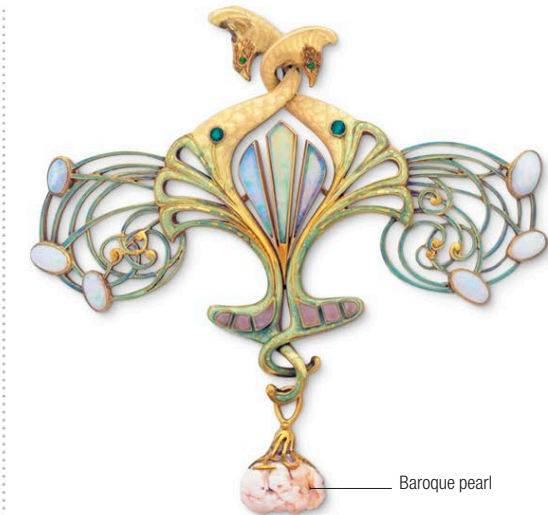


Locations
1 USA 2 Mexico 3 Honduras 4 Ethiopia 5 India
6 Australia 7 New Zealand

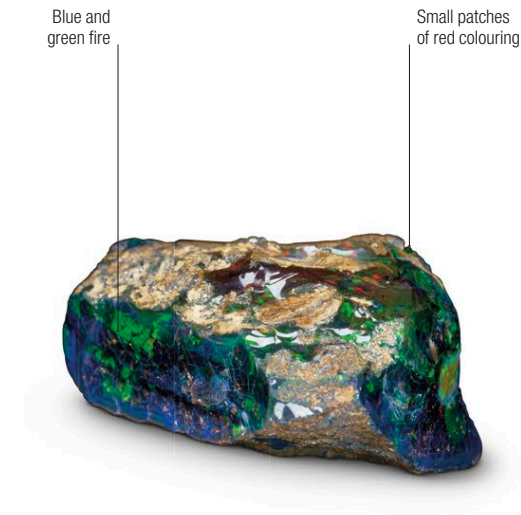
Key pieces



19th-century bracelet | Originating from Jaipur in India, this intricate gold bracelet is inlaid with fire opals, turquoises, and other precious and semi-precious stones. It also features enamelled decorative panels.



Peacock brooch | Designed by French jeweller Georges Fouquet, this gold brooch from around 1900 is decorated with opals, garnets, pearls, and enamel. Its delicate, almost organic form is typically Art Nouveau in style.



The Roebling Opal | Found in Virgin Valley, Nevada, USA, this huge, 2,585-carat piece of black opal rough exhibits vivid blue and green flashes of colour. Civil engineer John A. Roebling donated it to the Smithsonian's gem gallery in 1926.